DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

1 TITLE	2 IDENTIFICATION NUMBER
CUSTOM MICROCIRCUIT CHARACTERIZATION REPORT	DI-QCIC-80887

3 DESCRIPTION/PURPOSE

3.1 This report documents the data to be supplied in support of the requirements for the characterization of both custom integrated circuits and hybrid microcircuits.

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
890803	G/Y253		

7 APPLICATION/INTERRELATIONSHIP

7.1 This Data Item Description (DID) contains the format and content preparation instructions for the data product generated by the specific and discrete task requirement as delineated in the contract.

7.2 The term microcircuit or custom microcircuit, as used in this data item description, shall refer to both custom integrated circuits and hybrid microcircuits.

8 APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		G4797

10. PREPARATION INSTRUCTIONS

10.1 Format. The plan may be in the contractor's format.

10.2 Content. The report shall contain the following:

10.2.1 Title Page. The title page for the characterization report shall contain:

a. Contract number
b. Contractor's name
c. Equipment or purchase description identification
d. Date of preparation
e. Document title
f. Custom Microcircuits covered by the report
g. Security classification

10.2.2 Schmoo Plots. Schmoo plots shall be generated to chart the device operating region with respect to the following parameters at the devices specified lower temperature limit, 25 degree centigrade, and specified upper temperature limit. A minimum of 10 points along each parameter shall be chosen. Four points shall be in the specified operating region, such that the area will contain a total of 16 data points. Sample size shall be indicated. Data points which indicate failures shall be identified with a different character than those points which do not indicate failures.

7 DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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Block 10. PREPARATION INSTRUCTIONS (Continued)

a. Supply voltage vs. clock frequency
b. Clock pulse width (duty cycle vs. clock voltage)
c. Quiescent current vs. supply voltage
d. Dynamic current vs. supply voltage
e. Dynamic current vs. frequency

10.2.3 Discussion Section. This section shall describe the test setup and point out any important details and problems encountered.

10.2.4 Conclusion. The conclusion section shall give the contractor's final judgment as to whether the data presented has shown that the parts meet the specified operating requirements.

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